NOTATION · CHEAT SHEET

Derivative Notation at a Glance

The marks from the notation cards across the derivative pages: how each is read and what it means. Every group links to the full card on its page.

DERIVATIVES

$f'(x)$	**F PRIME OF X (LAGRANGE)** The derivative as a new function of x.
$\frac{dy}{dx}$	**D Y D X (LEIBNIZ)** Rate of change of y with x; one symbol, not a fraction.
$Df(x)$	**D F OF X (EULER)** D as an operator acting on f.
$\dot{y}$	**Y DOT (NEWTON)** Derivative with respect to time.

DERIVATIVE AS A FUNCTION

$\frac{f(x+h)-f(x)}{h}$	**THE DIFFERENCE QUOTIENT** Slope of a secant; its limit as $h \to 0$ is $f'(x)$.
$\left.\frac{dy}{dx}\right\|_{x=a}$	**D Y D X AT X EQUALS A** The derivative evaluated at one point: $f'(a)$.

DIFFERENTIABILITY

$f'_-(a),\ f'_+(a)$	**F PRIME MINUS, F PRIME PLUS AT A** One-sided derivatives; differentiable at a when they agree.
$f'(a) = \pm\infty$	**THE DERIVATIVE IS INFINITE** A vertical tangent: f is not differentiable there.

HIGHER-ORDER DERIVATIVES

$f'',\ f^{(n)}$	**F DOUBLE PRIME; F N-TH** Primes up to three, then the order in brackets.
$\frac{d^n y}{dx^n}$	**D N Y, D X TO THE N** Leibniz form of the n-th derivative.
$\ddot{y},\ D^n f$	**Y DOUBLE DOT; D TO THE N OF F** Second time derivative; D applied n times.

DIFFERENTIALS

$\Delta x,\ \Delta y$	**DELTA X, DELTA Y** Actual changes along the curve.
$dx,\ dy$	**DEE X, DEE Y** Changes along the tangent line: $dy = f'(x)\,dx$.

GRAPH ANALYSIS

c	**THE CRITICAL NUMBER C** Where $f'(c) = 0$ or $f'(c)$ does not exist.
$f' : + \to -$	**F PRIME CHANGES FROM POSITIVE TO NEGATIVE** Sign change of f': a local maximum.
$f' > 0$ on (a, b)	**F PRIME IS POSITIVE ON THE INTERVAL** f is increasing there.

TECHNIQUES

$2y\,\frac{dy}{dx}$	**TWO Y TIMES D Y D X** Implicit differentiation: the chain rule on y^2.
$(f^{-1})'(b)$	**DERIVATIVE OF F INVERSE AT B** Equals $\frac{1}{f'(a)}$ where $f(a) = b$.
$\frac{dy/dt}{dx/dt}$	**D Y D T OVER D X D T** The slope $\frac{dy}{dx}$ of a parametric curve.

SAME MARKS, DIFFERENT JOB

f'' vs f^2	The second derivative, not f squared; $f^{(2)}$ uses brackets for that reason.
$\frac{dy}{dx}$	One symbol, yet it behaves like a fraction in the chain rule and with differentials.
f^{-1}	The inverse function, not $\frac{1}{f}$.